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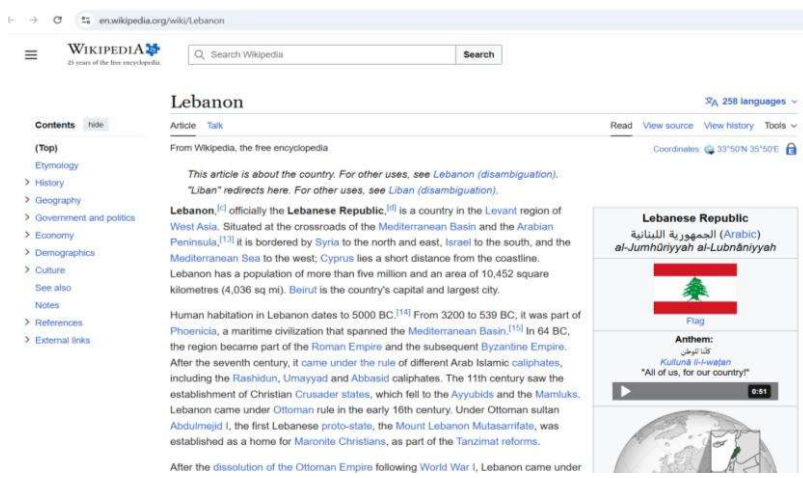
Project Proposal n#1

Wikipedia Semi-Structured Infobox Data Collection, Comparison, and Differencing

1. Title: Wikipedia Semi-Structured Infobox Data Collection, Comparison, and Differencing

2. Objective: The main objective of this project is to implement a semi-structured document comparison (tree edit distance) and differencing (patching) solution, which can be applied on different kinds of semi-structured documents consisting mainly of textual content. For the sake of the present project, we will specifically focus on comparing data extracted from Wikipedia, more specifically country-related infoboxes, for the purpose of comparing and differencing various countries against each other, attempting to understand their similarities and differences.



a. Sample Wikipedia page describing the country of Lebanon

Capital and largest city	Beirut 33°54'N 35°32'E
Official languages	Arabic[1]
Local vernacular	Lebanese Arabic[2]
Recognised minority language	French[3]
Ethnic groups (2021)[3]	95% Arab[3][4] 4% Armenian 1% others
Religion (2024)[5][6]	53.31% Islam — 26.84% Shia — 0.64% Alawite — 26.55% Sunni; 41.40% Christianity — 28.11% Catholic — 8.30% Orthodox — 2.92% Armenian Apostolic — 0.53% Protestant — 1.64% Other Christians; 5.21% Druze; 0.08% No religion
Demonym	Lebanese
Government	Unitary parliamentary republic under confessionalism[7]
• President	Joseph Aoun
• Prime Minister	Nawaf Salam
• Speaker of the Parliament	Nabih Berri
Legislature	Parliament

Establishment	
• Emirate of Mount Lebanon	1516
• Double Qaim-Maqamate of Mount Lebanon	1 December 1843
• Mount Lebanon Mutasarrifate	9 June 1861
• Greater Lebanon	1 September 1920
• Constitution	23 May 1926
• Independence declared	22 November 1943
• French mandate ended	24 October 1945
• Withdrawal of French forces	17 April 1946
• Israeli troops withdrawn	24 May 2000
• Syrian troops withdrawn	30 April 2005
Area	
• Total	10,452 km² (4,036 sq mi) (161st)
• Water (%)	1.8
Population	
• 2024 estimate	5,364,482[8] (117th)
• Density	513/km² (1,328.7/sq mi)
GDP (PPP)	2022 estimate
• Total	▲ $78.233 billion[9] (108th)
• Per capita	▲ $11,793[9] (114th)

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• Per capita	▲ $11,793[9] (114th)
GDP (nominal)	2022 estimate
• Total	▲ $21.780 billion[9] (103rd)
• Per capita	▲ $3,283[9] (133rd)
Gini (2011)	▼ 31.8[10] medium inequality
HDI (2023)	▲ 0.752[11] high (102nd)
Currency	Lebanese pound (LBP)
Time zone	UTC+2 (EET)
• Summer (DST)	UTC+3 (EEST)
Calling code	+961[3]
ISO 3166 code	LB
Internet TLD	.lb .لبنان

A map of Phoenicia, trade routes and the Phoenician colony of Carthage

b. Semi-structured data content from Wikipedia infobox

Figure 1. Sample Wikipedia page

3. Background and Motivations: Semi-structured XML (and more recently JSON) based data is increasingly available on the Web, due to the heterogeneity and diversity of data sources (e.g., scientific communities, industries, social media sites, and healthcare institutions) and the un-normalized nature of the data shared among those sources (e.g., publication records, reports, social and multimedia meta-data, and electronic health records). For instance, the Brazilian National Research Council (CNPq) requires every researcher to store their curricula in the Lattes platform[1], using XML as the representation format. The city of Baltimore in the USA annually publishes its financial, employment, security, healthcare, and other various records on line, represented in XML and JSON formats. Also, the scientific community performs regular data dumps regarding publications, authors, journals, and conferences[2]. Similarly, Wikipedia periodically generates dump files of the data it manages: articles, images, categories, constraints, and other metadata[3]. For instance, an extract of a sample XML document describing scientific publications from the Sigmod Record data repository[3] is shown in Figure 1.

Owing to their increasing Web presence, semi-structured document comparison and differencing becomes a central issue in the database and information retrieval communities, their applications ranging from: change management and data warehousing, data integration, search and retrieval, clustering and classification (cf. IDPA Ch. 4-5-6). In this context, Tree Edit Distance (*TED*, also known as *tree differencing*, *approximate tree matching*, or *tree-to-tree correction*) methods have been widely utilized to compare semi-structured data, and have been proven optimal w.r.t. less accurate structural comparison methods (cf. IDPA Ch. 4). *TED* techniques produce i) a similarity (distance) value, quantifying the resemblances/differences between two trees being compared, and ii) an edit script (also known as *diff* or *Delta*), describing the minimum amount of changes needed to transform one tree into another.

Hence, in this project, we propose to exploit the well know techniques for finding the edit distance between tree structures, to allow the comparison and differencing of semi-structured documents.

4. Project Description: The project can be divided in five main parts: i) data collection, ii) document pre-processing, iii) document tree comparison and differencing, iv) document tree patching, and v) document tree post-processing. Each component is described in more detail in the below paragraphs.

4.1. Data Collection

For the sake of the project, students are requested to build a special dataset: collecting data regarding all countries recognized by the United Nations[4], where each country is described using its own semi-structured document (cf. Figure 1.a). More specifically, country documents will be made of the semi-structured data contents collected from the right-hand side of the Wikipidia page describing the country (, i.e., the Wikipedia infobox, cf. Figure 1.b). Downloading the semi-structured Wikipedia infobox data can be achieved using several methods, ranging from high-level Python libraries to direct API calls. The infoboxes are usually formatted as HTML tables (class *infobox*), making them accessible for extraction. Once extracted, the data should be properly formatted as XML of JSON, for subsequent processing for TED computation and differencing. The collected data can be stored as native XML/JSON, or in dedicated XML/JSON (NoSQL) database tools, according to the student team preferences.

4.2. Data Pre-Processing

First, the tool should accept as input two semi-structured Wikipedia infobox documents (represented as XML/JSON, cf. Figure 2). It should then produce a *rooted ordered labeled tree* presentation of every document, in which the nodes represent semi-structured elements/attributes, labeled following element/attribute tag name. Element nodes are ordered following their order of appearance in the document. Attribute nodes appear as children of their encompassing element nodes, sorted left-to-right by attribute name, and appearing before sub-element siblings. Note that the order of attributes (unlike elements) is irrelevant in native XM/JSON, yet in the context of semi-structure comparison and processing, attribute nodes are usually ordered (as described above) so as to reduce the complexity of the similarity evaluation process. In this project, we also consider element/attribute values (*structure-and-content*) in the comparison process, where element/attribute values are mapped to leaf nodes labeled with their respective textual content (we will only consider textual content in this

[1] http://lattes.cnpq.br/
[2] https://dblp.uni-trier.de/xml/; http://www.dia.uniroma3.it/Araneus/Sigmod/; https://research.cs.wisc.edu/niagara/
[3] https://dumps.wikimedia.org/
[4] UN member states: https://en.wikipedia.org/wiki/Member_states_of_the_United_Nations

project, and disregarded other typed data like numbers, dates, Boolean, etc.). Textual values need to be *tokenized* (parsing names into tokens based on punctuation and case, e.g., sentence: "*I like data processing*" produces 4 tokens: '*I*', '*like*', '*data*', and '*processing*'), to be represented either: as one single text node in the document tree, or as separate nodes for every token, based on the students' choice of preference. The reason for making one choice or the other needs to be properly explained. *Stop word removal* and *stemming* will not be required at this stage (cf. Figure 3.a).

```
<?xml version='1.0' encoding='ISO-8859-1' ?>

<country>
    <name>Lebanon</name>
    <official_name>Lebanese Republic</official_name>

    <coordinates>
        <latitude>33°50'N</latitude>
        <longitude>35°50'E</longitude>
    </coordinates>

    <capital>Beirut</capital>
    <largest_city>Beirut</largest_city>

    <demonym>Lebanese</demonym>

    <government>
        <type>Unitary parliamentary republic under confessionalism</type>
        <president>Joseph Aoun</president>
        <prime_minister>Nawaf Salam</prime_minister>
        <speaker_of_parliament>Nabih Berri</speaker_of_parliament>
    </government>

    <legislature>Parliament</legislature>
    <area>
        <total_km2>10452</total_km2>
        <water_percent>1.8</water_percent>
    </area>

    <population>
        <estimate_year>2024</estimate_year>
        <total>5364482</total>
        <density_per_km2>513</density_per_km2>
    </population>

  …

</country>
```

Figure 2. Simplified extract of sample XML document from the Sigmod Record repository

4.3. Document Tree Similarity and Differencing

On the one hand, the similarity between two infobox document tree structures provides a very important piece of information in order to understand whether two countries are similar or different. A typical question we can ask ourselves here is: *How similar are Lebanon and Switzerland?* To compute the similarity between Lebanon and Switzerland infobox documents, tree edit distance (*TED*) has to be computed between the corresponding document trees, by adapting a *TED* algorithm provided in the literature (cf. IDPA, Ch. 5). The project should have a working implementation of the algorithm by Chawathe[5] or the algorithm by Nierman and Jagadish's[6], or another similar TED algorithm. Note that *TED* computations need to distinguish between document structure (element/attribute names) and content (element/attribute values).

On the other hand, the differencing between two infobox documents will describe how the two countries being compared are similar or different. In other words, the question we can ask ourselves here is: *How are Lebanon and Switzerland similar?* To identify the difference between two infobox documents, the edit script or *diff* between the two compared trees needs to be computed. The *diff* computation process consists of a backtracking of the TED computation described in the above paragraph, and will depend on the nature of the TED algorithm adopted (cf. Figure 3.b).

[5] Chawathe S., Comparing Hierarchical Data in External Memory. International Conference on Very Large Data Bases (VLDB), 1999. pp. 90-101.

[6] Nierman A. and Jagadish H. V., Evaluating structural similarity in XML documents. Proceedings of the ACM SIGMOD International Workshop on the Web and Databases (WebDB), 2002. pp. 61-66.

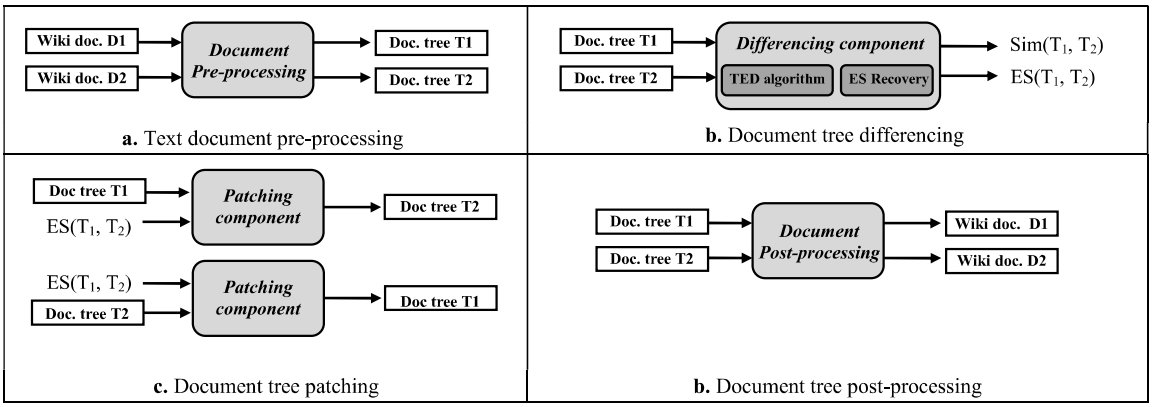

Figure 3. Semi-structured document differencing and patching

Having identified the edit script, otherwise known as *diff*, describing the changes between two document trees (T_1 and T_2) compared via the *TED* algorithm, the *diff* would have to be outputted in a useful format to be efficiently exploited in the remainder of the study. For instance, an edit script ES(T_1, T_2) = $\prec$ *Ins*(T_1[2], 1, T_2[2]), *Upd*(T_1[3],'*G*') $\succ$ should be transformed and represented in a readable way (XML/JSON) to be stored in a separate document for later usage in patching. Here, the students can come up with their own syntax to describe *diffs*, or they can adapt existing *diff* representations such as DeltaXML[7] [9] and XyDiff[8] [10]. In the context of this project, the *diff* between two Wikipedia infobox documents will describe the difference between the corresponding countries (e.g., difference between Lebanon and Switzerland in geographic size, population size and diversity, GDP, etc.).

4.4. Document Tree Patching

After identifying the edit script (*diff*) transforming one document tree into another, a complementary functionality would be to apply the *diff* to either one of the source trees, in order to obtain the other tree (Figure 3.c). This action is known as *patching*. In the context of this specific project, patching will allow us to understand what needs to be done to transform one country into the other. In other works, we ask ourselves here: *Given the similarity between Lebanon and Switzerland (established in the previous TED step), what would it take to transform Lebanon into Switzerland if we wanted to?*

While various tools have been developed for *TED* computations (*differencing*), very few implementations enable *patching*. A main aspect here is the *diff* output format itself, and how formatted *diffs* should be handled in order to re-integrate the changes in the original trees.

4.5. Document Tree Post-Processing

Finally, the resulting trees need to be post-processed (Figure 3.d) to be represented in their native XML/JSON format and eventually in their original Wikipedia infobox format.

5. Expected Results: The main objective of this project is to develop and implement the Wikipedia semi-structured infoboxes document pre-processing, comparison, differencing, and patching modules described above. Each student group is expected to conceive and implement:

1. A pre-processing module to transform input Wikipedia infoboxe documents into rooted ordered labeled trees,
2. A *TED* algorithm, which takes as input two infobox trees, and outputs the distance (similarity) value,
3. An edit script extraction algorithm, which extract the edit script describing the transformation operations between two infobox trees, based on their computation,
4. A tree patching algorithm, which takes as input the document tree and edit script, and applies the edit script to the input tree in order to obtain the other tree,
5. A post-processing tool to transform the document tree into its original Wikipedia infobox textual form.

[7] Monsell EDM ltd. *DeltaXML*. http://deltaxml.com
[8] http://wwwrocq.inria.fr/~cobena/XyDiffWeb/.